

```

1 package jdbcExersice;
2 import java.sql.Connection;
3 import java.sql.DriverManager;
4 import java.sql.PreparedStatement;
5 import java.sql.ResultSet;
6 import java.sql.SQLException;
7 import java.util.Scanner;
8
9 public class DisplayCountries {
10
11     public static void main(String[] args) {
12
13         String url = "jdbc:mariadb://localhost:3306/nation";
14         String user = "user";
15         String password = "easwaren123";
16
17         Scanner input = new Scanner(System.in);
18         System.out.print("Enter a language name (e.g., English, French): ");
19         String targetLanguage = input.nextLine();
20
21         try {
22             Connection conn = DriverManager.getConnection(url, user, password);
23             System.out.println("Connection to MariaDB established successfully!");
24             System.out.println("\nCountries where " + targetLanguage + " is spoken:");
25             System.out.println("-----");
26
27             // Added 'cl.official' to our SELECT statement
28             String query = "SELECT c.name AS CountryName, cl.official AS IsOfficial " +
29                 "FROM countries c " +
30                 "JOIN country_languages cl ON c.country_id = cl.country_id " +
31                 "JOIN languages l ON cl.language_id = l.language_id " +
32                 "WHERE l.language = ?";
33
34             PreparedStatement pstmt = conn.prepareStatement(query);
35             pstmt.setString(1, targetLanguage);
36

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33
34 PreparedStatement pstmt = conn.prepareStatement(query);
35 pstmt.setString(1, targetLanguage);
36
37 ResultSet rs = pstmt.executeQuery();
38
39 boolean found = false;
40
41 while (rs.next()) {
42     String countryName = rs.getString("CountryName");
43
44     // Get the official status (JDBC reads 1/0 or true/false natively as a boolean)
45     boolean isOfficial = rs.getBoolean("IsOfficial");
46
47     // Format the output based on that boolean
48     String statusText;
49     if (isOfficial) {
50         statusText = "Official";
51     } else {
52         statusText = "Unofficial";
53     }
54
55     // Print the country name alongside its status
56     System.out.println("- " + countryName + " (" + statusText + ")");
57     found = true;
58 }
59
60 if (!found) {
61     System.out.println("No countries found in the database for that language.");
62 }
63
64 rs.close();
65 pstmt.close();
66 conn.close();
67 input.close();
68
69 } catch (SQLException e) {
70     System.err.println("Database connection failed:");
71     e.printStackTrace();
72 }
73 }
74 }

```

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59
60     if (!found) {
61         System.out.println("No countries found in the database for that language.");
62     }
63
64     rs.close();
65     pstmt.close();
66 }
```

Problems Javadoc Declaration Console × Terminal Coverage

<terminated> DisplayCountries [Java Application] C:\Users\bloop\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.8.v20250724-1412

Enter a language name (e.g., English, French): english

Connection to MariaDB established successfully!

Countries where english is spoken:

```
-----
- Aruba (Unofficial)
- Anguilla (Official)
- Netherlands Antilles (Unofficial)
- American Samoa (Official)
- Antigua and Barbuda (Official)
- Australia (Official)
- Bahrain (Unofficial)
- Belize (Official)
- Bermuda (Official)
- Barbados (Official)
- Brunei (Unofficial)
- Canada (Official)
```

- Canada (Official)
- Cocos (Keeling) Islands (Official)
- Cook Islands (Unofficial)
- Christmas Island (Official)
- Cayman Islands (Official)
- Denmark (Unofficial)
- Falkland Islands (Official)
- United Kingdom (Official)
- Gibraltar (Official)
- Guam (Official)
- Hong Kong (Official)
- Ireland (Official)
- Iceland (Unofficial)
- Japan (Unofficial)
- Saint Kitts and Nevis (Official)
- Kuwait (Unofficial)
- Saint Lucia (Official)
- Lesotho (Official)
- Macao (Unofficial)
- Monaco (Unofficial)
- Maldives (Unofficial)
- Marshall Islands (Official)
- Malta (Official)
- Northern Mariana Islands (Official)
- Montserrat (Official)
- Malaysia (Unofficial)
- Norfolk Island (Official)
- Niue (Official)
- Norway (Unofficial)
- Nauru (Official)
- New Zealand (Official)
- Palau (Official)
- Puerto Rico (Unofficial)
- Saint Helena (Official)
- Seychelles (Official)
- Turks and Caicos Islands (Official)
- Tokelau (Official)
- Tonga (Official)
- Trinidad and Tobago (Unofficial)
- Tuvalu (Official)
- United States Minor Outlying Islands (Official)
- United States (Official)

- United States (Official)
- Saint Vincent and the Grenadines (Official)
- Virgin Islands, British (Official)
- Virgin Islands, U.S. (Official)
- Vanuatu (Official)
- Samoa (Official)
- South Africa (Official)
- Zimbabwe (Official)